

Name: **Maths Class:**

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 2 Mathematics

TRIAL HIGHER SCHOOL CERTIFICATE

August 2009

TIME ALLOWED: 120 minutes

READING TIME: 5 minutes

Instructions:

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- All questions are of equal value.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- **START ALL QUESTIONS ON A NEW PAGE**
- Approved calculators may be used.
- A table of *Standard Integrals* is attached. You may detach this page now.

(FOR MARKERS USE ONLY)

[illegible]

QUESTION 1:

Marks

(a) Find

2 (i) $\int \cos^3 x \, dx$

2 (ii) $\int \frac{dx}{x^2 - 4x + 8}$

2 (iii) $\int_1^5 \frac{dx}{(2x-1)\sqrt{2x-1}}$

4 (b) Prove that $\sec x = \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x}$

and hence show that $\int_0^{\frac{\pi}{4}} \sec x \, dx = \ln(\sqrt{2} + 1)$

5 (c) (i) Find values of A , B and C so that

$$\frac{5}{(x^2+4)(x+1)} = \frac{Ax+B}{x^2+4} + \frac{C}{x+1}$$

(ii) Hence find $\int \frac{5 \, dx}{(x^2+4)(x+1)}$

QUESTION 2: (Start a new page)

Marks

- 6** (a) If $z = 1 - i$, find
- (i) $\bar{z}$ (ii) $|z|$ (iii) $\arg z$ (iv) $\arg iz$ (v) z^6 (in simplest form)

- 2** (b) (i) Sketch the region where the inequalities

$$|z - 2| \leq |z - 2i| \quad \text{and} \quad |z - 1 - 2i| \leq 1$$

hold simultaneously.

- 3** (ii) P is a point on the boundary of the region in part (i) above, and is represented by the complex number z , where $\arg z = \frac{\pi}{4}$.

Find the 2 possibilities for z (in the form $a+ib$).

- 4** (c) A plane curve is defined by the equation

$$x^2 + 2xy + y^5 = 4$$

The curve has a horizontal tangent at the point $P(X, Y)$.

By using implicit differentiation, or otherwise, show that X is the unique solution to

$$X^5 + X^2 + 4 = 0$$

QUESTION 3: (Start a new page)**Marks**

2 (a) (i) Without using calculus, sketch the curve $y = (x + 1)^2(1 - x)$

2 (ii) On a separate diagram from above, but using the same scale on the axes, *and also without calculus*, sketch the curve

$$y^2 = (x + 1)^2(1 - x)$$

In your answer, pay close attention to the shape of the curve as y approaches zero.

6 (b) Sketch each of the following curves on separate axes for $0 \leq x \leq 2\pi$

(i) $y = \sin^2 x$

(ii) $y = |\sin x|$

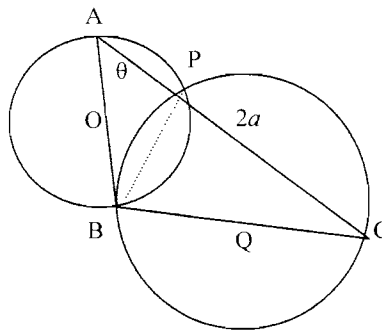
(iii) $y = \sqrt{\sin^2 x}$

(iv) $y = \frac{1}{\sin x}$

(v) $y = \frac{|\sin x|}{\sin x}$

(vi) $y = e^{\sin x}$

(c) The hypotenuse AC of a right-angled triangle ABC has a length of $2a$ units and makes an angle of θ with one of the shorter sides, as shown below.



Circles are drawn using the two shorter sides as diameters, intersecting at points B and P. For this diagram, P is NOT on the side AC.

O and Q are the centres of the circles.

(i) Redraw the diagram in your answer book. (No marks)

2 (ii) Prove that the point P lies on AC (you may initially assume that it doesn't)

3 (iii) Show that the length of PB is $a \sin 2\theta$

QUESTION 4: (Start a new page)

Marks

2 (a) Show that $\int_0^{\frac{\pi}{4}} \tan \theta d\theta = \frac{1}{2} \ln 2$

2 (b) (i) Prove that, for any complex numbers z_1 and z_2

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

3 (ii) Hence, using the method of Mathematical Induction, prove that

$$\arg(z_1 z_2 \dots \dots z_n) = \arg z_1 + \arg z_2 + \dots \dots \dots + \arg z_n$$

(c) A cubic polynomial is given by $P(x) = x^3 + ax + b$

where a and b are constants.

It is given that the polynomial equation $P(x) = 0$ has three roots, α , β , and γ

1 (i) Find the value of $\alpha + \beta + \gamma$

2 (ii) Show that $\alpha^2 + \beta^2 + \gamma^2 = -2a$

3 (iii) If the polynomial has a double root, show that this double root is $\frac{-3b}{2a}$

2 (iv) If the polynomial has 3 distinct roots, show that $4a^3 + 27b^2 < 0$

QUESTION 5: (Start a new page)

Marks

5 (a) Given the hyperbola $16x^2 - 9y^2 = 144$, find

(i) the length of the major axis

(ii) the eccentricity

(iii) the co-ordinates of the foci

(iv) the equations of the directrices

(v) the equations of the asymptotes

(b) The parametric co-ordinates of a point P on the curve $y^2 = x^3$ are $x = t^2$ and $y = t^3$

2 (i) Show that the equation of the tangent to this curve at P is

$$t^3 - 3tx + 2y = 0$$

1 (ii) Explain why there can be no more than 3 distinct tangents to $y^2 = x^3$ drawn from any remote point (x_1, y_1) , which is not on the curve.

2 (iii) Show that if the tangents to the curve at the points on it having parameters t_1 , t_2 and t_3 all pass through the remote point (x_1, y_1) , then

$$t_1^2 + t_2^2 + t_3^2 = 6x_1$$

5 (c) The area under the curve $y = x^2$, above the x -axis and between the lines $x = 1$ and $x = 2$, is rotated through 2π radians about the line $x = 2$.

Using the method of cylindrical shells, show that the volume of the solid so formed is $\frac{11\pi}{6}$ cubic units.

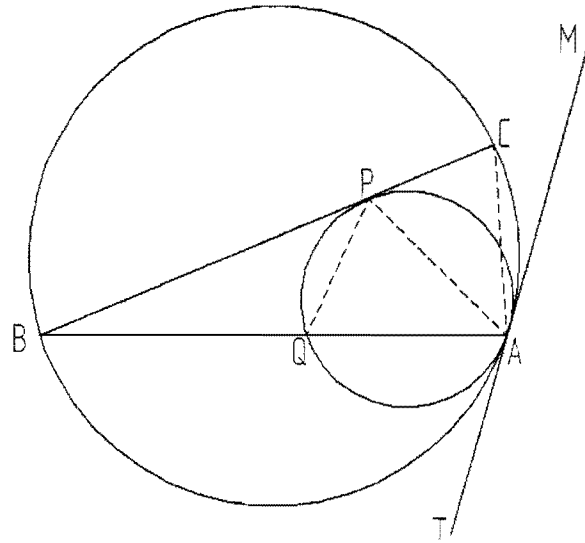
QUESTION 6: (Start a new page)

Marks

- (a) Two circles touch internally at a point A and have a common tangent TAM as shown below.

A tangent to the inner circle through a point P (which is not the centre of either circle) meets the outer circle at B and C.

AB cuts the inner circle at Q.



- (i) Redraw the diagram neatly onto your answer page (*no marks*).

5

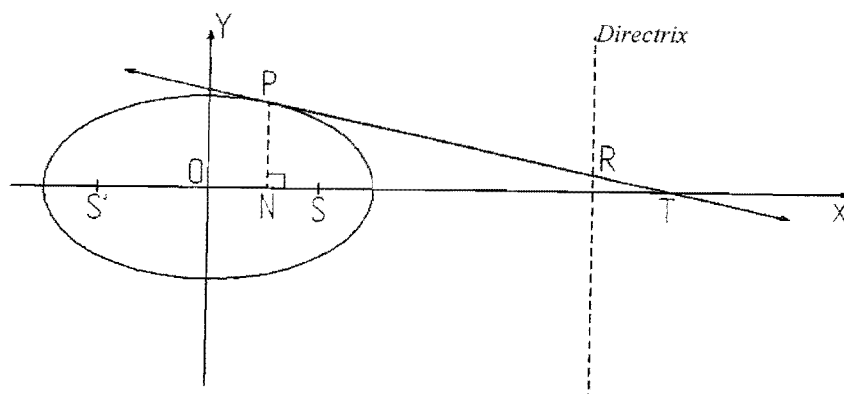
- (ii) Giving all appropriate reasons, prove that AP bisects the angle BAC.

QUESTION 6 continues over the page.....)

QUESTION 6 continued.....)

(b)

$P(a\cos\theta, b\sin\theta)$ is any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



The tangent at P cuts the major axis of the ellipse at T and the Directrix at R, while N is the foot of the perpendicular from P to the x-axis.

O is the centre of the ellipse, while S and S' are the foci.

3

- (i) Show that the equation of the tangent at P is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$
(Show all working)

2

- (ii) Show that $ON \cdot OT = a^2$

5

- (iii) Showing all steps carefully, prove that PR subtends a right angle at S.

QUESTION 7: (Start a new page)

Marks

3 (a) Using the substitution $x = a \tan \theta$, or otherwise, find $\int \frac{dx}{(a^2 + x^2)^{\frac{3}{2}}}$

(b) You are given the complex polynomial $P(z) = z^5 - 1$

The roots of $P(z) = 0$ are $1, \omega_1, \omega_2, \omega_3, \omega_4$ which are in cyclic order around the unit circle.

3 (i) Prove the following:

(α) $\omega_1 = \overline{\omega_4}$ and $\omega_2 = \overline{\omega_3}$

(β) $\omega_1 + \omega_2 + \omega_3 + \omega_4 = -1$

(γ) $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$

2 (ii) Using the sum of the products of the roots taken in pairs, or otherwise, show that

$$4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} + 1 = 0$$

1 (iii) Deduce that $\cos \frac{2\pi}{5}$ and $\cos \frac{4\pi}{5}$ are solutions to $4x^2 + 2x - 1 = 0$

4 (c) (i) If $I_n = \int_0^{\frac{\pi}{4}} \sec^n \theta d\theta$,

show that $(n-1)I_n - (n-2)I_{n-2} = (\sqrt{2})^{n-2}$, for $n \geq 2$

2 (ii) Using part (i) above, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 \theta d\theta$

QUESTION 8: (Start a new page)

Marks

- (a) In the right triangular prism shown,

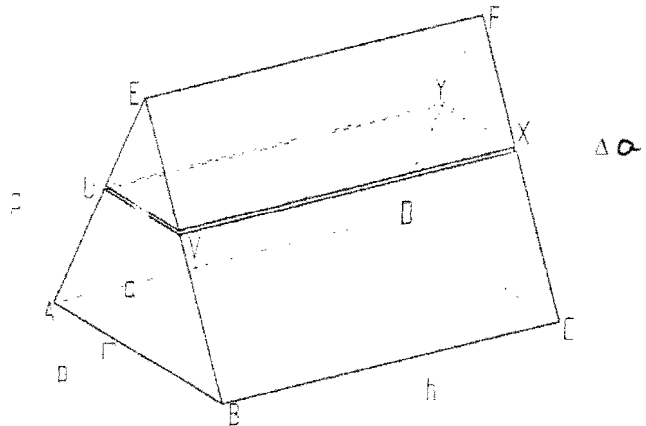
$$AB=DC= b \text{ units}$$

$$AE=BE=DF=CF$$

M is the midpoint of AB

$$EM = p \text{ units}$$

$$BC=AD=EF= h \text{ units}$$



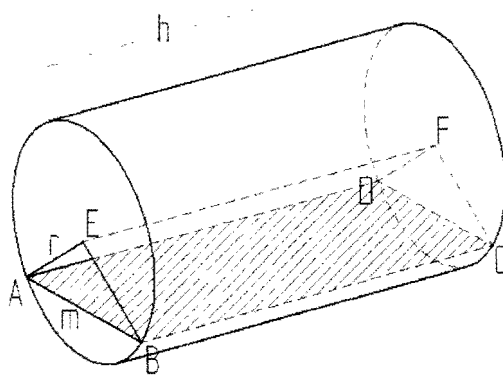
A “slice” UVXY of thickness Δa is taken a units above the base ABCD and parallel to it.

- 3 (i) Show that the volume of the rectangular slice is given by

$$\Delta V = \left(\frac{p-a}{p} \right) bh \Delta a$$

- 2 (ii) Hence, show that the volume of the triangular prism is given by $V = \frac{1}{2}pbh$

- 4 (iii) The triangular prism above is fitted into a right circular cylinder, of base radius r units and height h units, as shown below, where the points E and F are the centres of the circular bases.



Taking the angle AEB as $\frac{2\pi}{n}$, verify that the volume of the cylinder is $\pi r^2 h$
 (In your proof you may use the result $\lim_{x \rightarrow 0} \tan x = x$)

QUESTION 8 continues over.....)

QUESTION 8 continued.....)

- 6** (b) A particle P moves in the x,y -plane and its co-ordinates (x, y) satisfy the equations

$$\frac{d^2x}{dt^2} = -n^2x \quad \text{and} \quad \frac{d^2y}{dt^2} = -n^2y, \quad \text{where } n \text{ is a constant}$$

Initially ($t=0$), it is given that $x = 4$, $y = 0$, $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 3n$

Show that, as t varies, x and y describe the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$

END OF EXAMINATION PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Teacher's Name:

Student's Name/N^o:SOLUTIONS AND MARKINGQUESTION 1:

$$(a)(i) \int \cos^3 x \, dx = \int \cos x (1 - \sin^2 x) \, dx$$

$$= \int \cos x \, dx - \int \cos x \sin^2 x \, dx$$

2 MARKS

1 for bracking it up

$$= \sin x - \frac{1}{3} \sin^3 x + h$$

1 for answer.

$$(ii) \int \frac{dx}{x^2 - 4x + 8} = \int \frac{dx}{(x-2)^2 + 4}$$

2 MARKS

Award

1 for either completing the square or for tan.

$$= \frac{1}{2} \tan^{-1} \frac{x-2}{2} + h$$

$$(iii) \int_1^5 \frac{dx}{(2x-1)\sqrt{2x-1}} = \int_1^5 (2x-1)^{-3/2} dx$$

Do NOT PENALISE for no h.

2 MARKS

$$= \left[- (2x-1)^{-1/2} \right]_1^5$$

$$= \left[\text{OR } - \frac{1}{\sqrt{2x-1}} \right]_1^5$$

$$= -\frac{5}{\sqrt{9}} + \frac{1}{\sqrt{1}}$$

$$= -\frac{2}{3}$$

either earns the first mark.

Second mark

$$(b) \text{ RHS} = \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x}$$

1 MARK.

$$= \sec x = \text{LHS.}$$

$$\therefore \int_0^{\pi/4} \sec x \, dx = \int_0^{\pi/4} \frac{\sec x \tan x + \sec x}{\sec x + \tan x} dx$$

1 MARK for

using part (i)

$$= \ln (\sec x + \tan x) \Big|_0^{\pi/4}$$

1 MARK for recognising this.

$$= \ln (\sqrt{2} + 1) - \ln 1$$

$$= \ln (\sqrt{2} + 1)$$

1 MARK

Teacher's Name:

Student's Name/Nº:

Q. d)

$$C = 1$$

$$(Ax+B)(x+1) + C(x^2+4) = 5$$

$$\rightarrow A = -1 \text{ and } A+B=0 \Rightarrow B=1$$

$$\therefore \int \frac{5 \, dx}{(x^2+4)(x+1)} = \int \frac{1-x}{x^2+4} \, dx + \int \frac{dx}{x+1}$$

$$= \int \frac{dx}{x^2+4} - \frac{1}{2} \int \frac{2x}{x^2+4} \, dx + \ln(x+1) + k$$

$$= \frac{1}{2} \tan^{-1} \frac{x}{2} - \frac{1}{2} \ln(x^2+4) + \ln(x+1) + k$$

2 marks for

A, B, C.

no matter how!

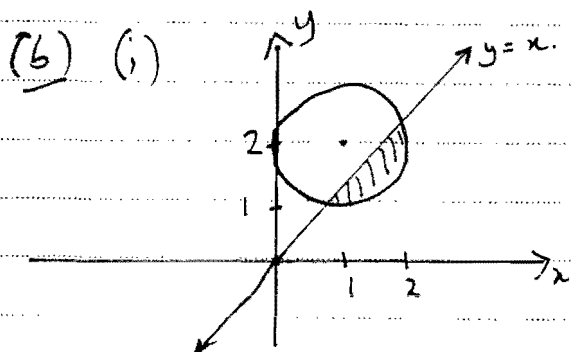
1 mark for
each part.

QUESTION 2:

(a) (i) $z = 1 - i$ (ii) $|z| = \sqrt{2}$
 $\bar{z} = 1 + i$

(iii) $\arg z = \begin{cases} \pi/4 \\ \text{or } -\pi/4 \end{cases}$ (iv) $\arg iz = \pi/4$

(v) $z = \sqrt{2} \operatorname{cis}(-\pi/4)$
 $z^6 = 8 \operatorname{cis}(-3\pi/2)$
 $= 8 \operatorname{cis}(\pi/2)$
 $= 8i$



(ii) P lies on $y = x$ as $\arg z = \pi/4$

$\therefore$ Solving simultaneously

$y = x$ and $(x^2 - 1)^2 + (y - 2)^2 = 1$

gives $2x^2 - 6x + 5 = 1$

$x^2 - 3x + 2 = 0$

$(x - 2)(x - 1) = 0$

$\therefore x = 2$ or $x = 1$

$y = 2$ $y = 1$

$\therefore P$ is $2 + 2i$ or $1 + i$

(c) $2x + 2y + 2x \frac{dy}{dx} + 5y + \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} (2x + 5y + 1) = -2(x + y)$

$\frac{dy}{dx} = \frac{-2(x + y)}{2x + 5y + 1}$

Since there is a horizontal tangent at (X, Y) , $\frac{dy}{dx} = 0$

$\therefore -2(X + Y) = 0$

$\therefore Y = -X$

6 MARKS

1 for each part
 (i) to (v)

2 marks for part (v)
 (1 ONLY for $8 \operatorname{cis} \pi/2$)

2 MARKS

1 for each of the
 areas inside circle and
 below $y = x$

3 MARKS

1 for recognising this

1 for this step

note
 3 for P no
 matter how

1 for this

4 MARKS

1 mark

1 for this

1 for this

Teacher's Name:

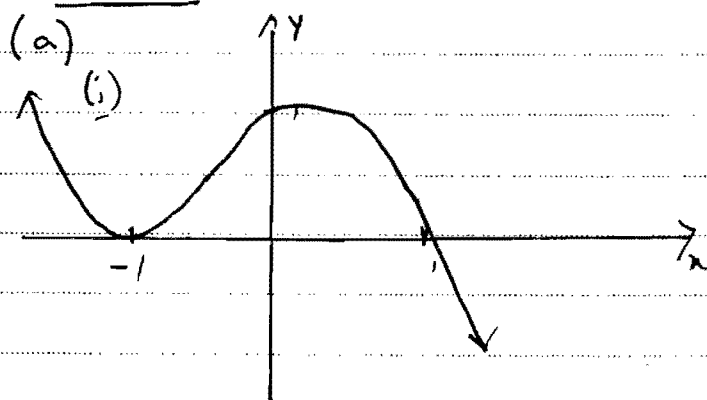
Student's Name/N^o:

∴ Equation becomes

$$x^2 + 2x(-x) + (-x)^5 = 4$$

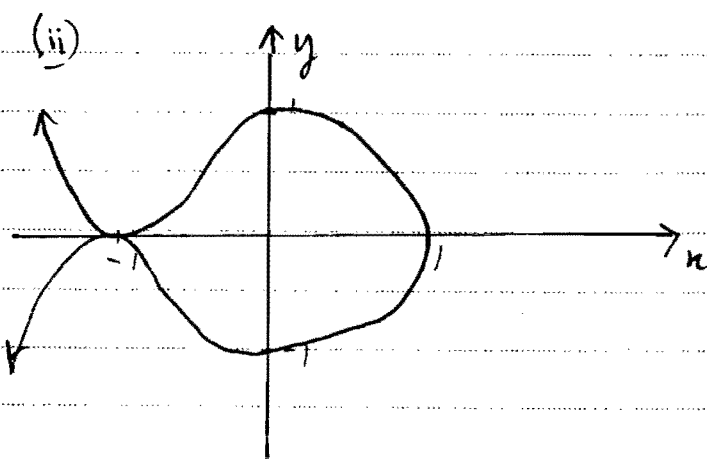
$$\therefore x^5 + x^2 + 4 = 0$$

← 1 for final statement

QUESTION 3:2 MARKS

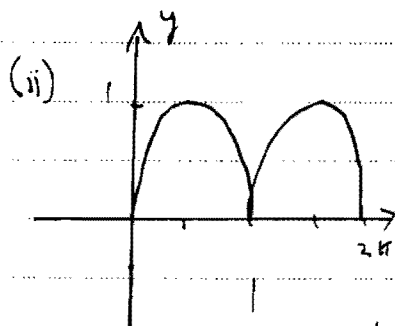
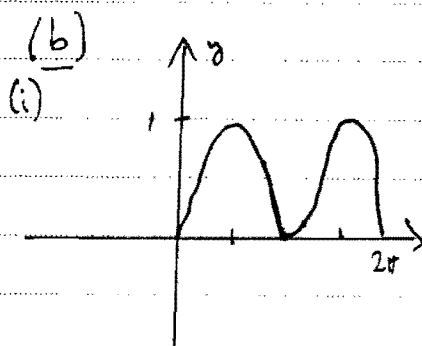
Key features are:

- cutting axis at $(1, 0)$
 - bouncing off axis at $(-1, 0)$
 - Right way up for a negative cubic
- [NO PENALTY for not having $(0, 1)$]

2 MARKS

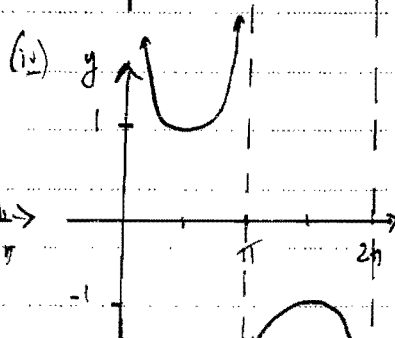
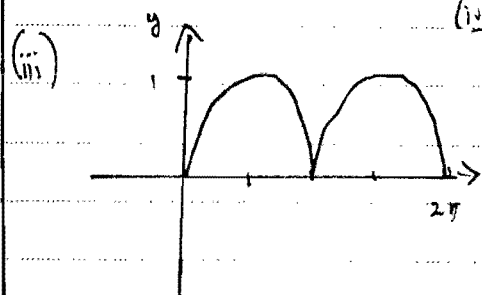
Key features are:

- Bounces at $(-1, 0)$
 - "Rounded" at $(1, 0)$
- 1 mark for having both positive and negative parts.
- [SUBTRACT 1 MARK if the graph goes outside $x \geq 1$]

1 EACH (Key points are:)

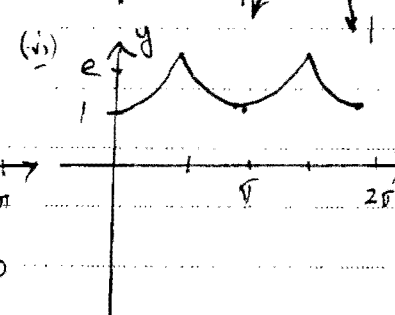
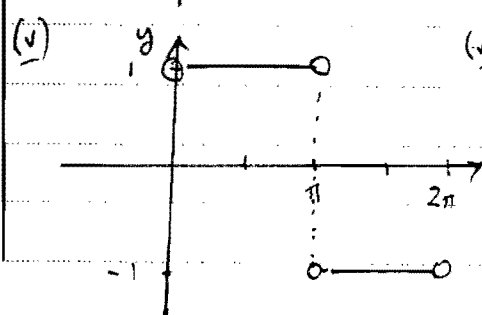
(i) no pointy bits - rounded

(ii) NOT rounded - pointy



(iii) must be all positive.

(iv) Asymptotes to be shown or be obvious

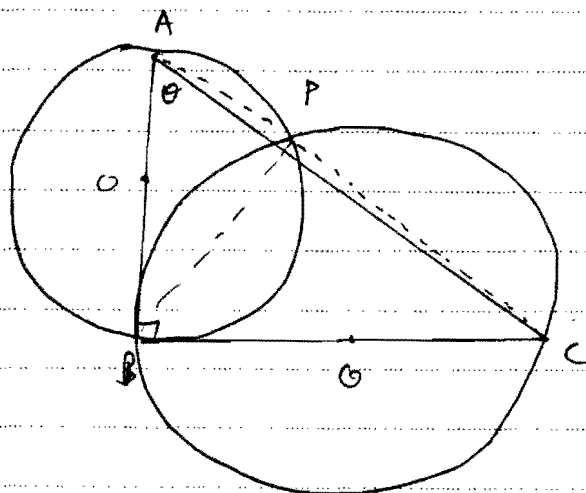


(v) Important to show open circles at endpoints

(vi) not necessary to show e . Shape is important

3 cons. . . .)

(c)



(ii) By joining AP and PC,

In smaller circle, AB is a diameter

 $\therefore \angle APB = 90^\circ$ (angle in a semi-circle)Similarly in the larger circle $\angle APB = 90^\circ$ $\therefore \angle APC = 180^\circ$ $\therefore P$ lies on AC.

(i) 2 MARKS

They have to convince you.

Do NOT accept things like "obvious".

(iii) In $\triangle ABC$, $\frac{BC}{AC} = \sin \theta$

$$\therefore BC = 2a \sin \theta$$

In $\triangle BPC$, $\angle PCB = (90 - \theta)^\circ$ (angle sum of $\triangle BPC$)

$$\therefore \frac{PB}{BC} = \sin(90 - \theta)$$

$$\therefore PB = BC \cos \theta$$

$$= 2a \sin \theta \cos \theta$$

$$= a \sin 2\theta$$

3 MARKS

1 MARK

1 MARK

1 MARK

OR

In $\triangle ABC$, $\frac{AB}{AC} = \cos \theta$

$$\therefore AB = AC \cos \theta$$

$$= 2a \cos \theta$$

In $\triangle APB$, $\frac{PB}{AB} = \sin \theta$

$$\therefore PB = AB \sin \theta$$

$$= 2a \cos \theta \sin \theta$$

$$= a \sin 2\theta$$

1 MARK

1 MARK

1 MARK

QUESTION 4:

$$\begin{aligned}
 (a) \int_0^{\pi/4} \tan \theta d\theta &= \int_0^{\pi/4} \frac{\sin \theta}{\cos \theta} d\theta \\
 &= -\ln \cos \theta \Big|_0^{\pi/4} \\
 &= -\ln \frac{1}{\sqrt{2}} + \ln 1 \\
 &= \frac{1}{2} \ln 2
 \end{aligned}$$

2 MARKS

1 for this line

1 for this

(b) (i) let $z_1 = r_1 \cos \theta_1$, and $z_2 = r_2 \cos \theta_2$

$$\therefore z_1 z_2 = r_1 r_2 \cos \theta_1 \cos \theta_2$$

$$= r_1 r_2 [\cos(\theta_1 + i \sin \theta_1)] [\cos \theta_2 + i \sin \theta_2]$$

$$= r_1 r_2 \left(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 + i (\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \right)$$

$$= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

$$= r_1 r_2 \cos(\theta_1 + \theta_2)$$

$$\therefore \arg z_1 z_2 = \arg z_1 + \arg z_2$$

2 MARKS

(ii) For $n=2$ the formula is true (above)Assume the formula is true for $n=k$

$$\text{ie } \arg(z_1 z_2 \dots z_k) = \arg z_1 + \arg z_2 + \dots + \arg z_k$$

For $n=k+1$

$$\arg(z_1 z_2 \dots z_k z_{k+1}) = \arg[(z_1 \dots z_k) z_{k+1}]$$

← 1 MARK

$$= \arg(z_1 \dots z_k) + \arg z_{k+1}$$

(from part (i))

$$= \underbrace{\arg z_1 + \arg z_2 + \dots + \arg z_k}_{\text{from assumption}} + \arg z_{k+1}$$

← 1 MARK

∴ If the formula is true for $n=k$, it is true for $n=k+1$

But it is true for $n=2$

∴ " " " " $n=3$ and so on.

ie the formula is true for all n .

1 MARK

Q 4 con...

(c) (i) $\alpha + \beta + \gamma = 0$

1 MARK

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$

1 MARK

$$= -2a$$

1 MARK

(iii) If there is a double root, it solves $P'(x) = 0$

3 MARKS

ie $3x^2 + a = 0$

$$\therefore x = \pm \sqrt{-a/3}$$

1 MARK

$$\therefore P(\sqrt{-a/3}) = 0 \Rightarrow \left(-\frac{a}{3}\right)^{3/2} + a(\sqrt{-a/3}) + b = 0$$

$$\therefore \left(-\frac{a}{3}\right)^{1/2} \left[\left(-\frac{a}{3}\right) + a \right] + b = 0$$

$$\therefore \left(-\frac{a}{3}\right)^{1/2} = -b/2a/3$$

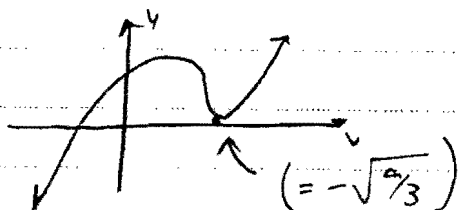
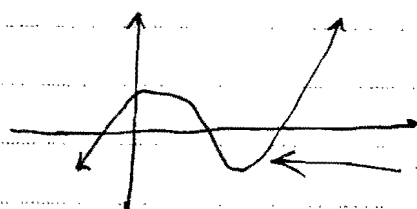
$$= -3b/2a$$

$$\therefore \text{double root is } -3b/2a$$

2 MARKS

(iv) If this polynomial has a double root its graph looks like:

2 MARKS

For 3 distinct roots, this becomesfor this area $P(x) < 0$

ie. $P(\sqrt{-a/3}) < 0 \Rightarrow \left(-\frac{a}{3}\right)^{3/2} < -\frac{3b}{2a}$

$$\therefore -a/3 > 9b^2/4a^2$$

$$\therefore -4a^3 > 27b^2$$

$$27b^2 + 4a^3 < 0$$

This is the key to it
= 1 MARK

1 MARK for algebra

(WATCH negative 'judges')

QUESTION 5:

(a) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(i) length of axis = 6

(ii) $16 = 9(e^2 - 1)$

$\therefore e^2 = 1 + \frac{16}{9}$

$e = \frac{5}{3}$

(iii) foci at $(\pm 5, 0)$

(iv) Directrices at $x = \pm \frac{9}{5}$

(v) Asymptotes are $y = \pm \frac{4}{3}x$

(5 MARKS)

1 each

(b) (i) P is (t^2, t^3)

$2y \frac{dy}{dx} = 3x^2$

$\therefore \frac{dy}{dx} = \frac{3x^2}{2y}$

At P, $m_T = \frac{3t}{2}$

Equation of tangent is:

$y - t^3 = \frac{3t}{2}(x - t^2)$

$\therefore t^3 - 3tx + 2y = 0$

2 MARKS

1 for slope

1 for equation

(ii) The parameters of any point P (not on the curve) which has a tangent to the curve at (x, y) solve $t^3 - 3tx + 2y = 0$ and this has at most 3 solutions for t i.e. there are no more than 3 tangents

1 for seeing this connection

(iii) If there are 3 points, then their parameters

2 MARKS

t, t_1, t_2 are solutions to $t^3 - 3tx + 2y = 0$

where $x = x$, and $y = y$,

Sum of these = $t + t_1 + t_2 = 0$

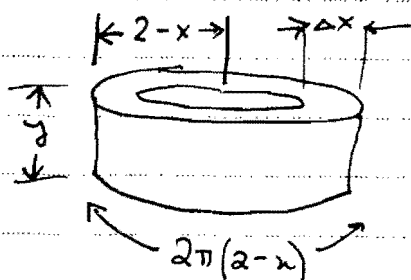
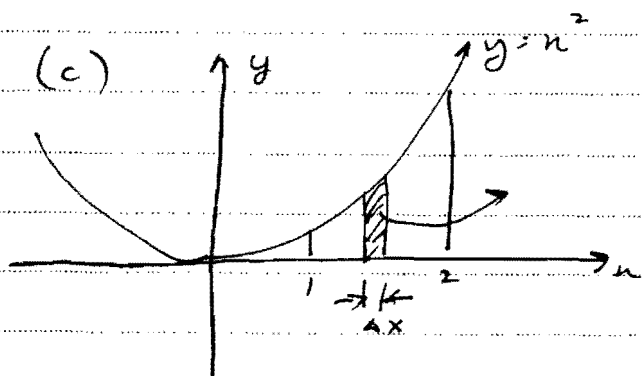
Product in pairs = $t \cdot t_1 + t \cdot t_2 + t_1 \cdot t_2 = -3x$, ← 1 for all this

Since $t^2 + t_1^2 + t_2^2 = (t + t_1 + t_2)^2 - 2(t \cdot t_1 + t \cdot t_2 + t_1 \cdot t_2)$

$= 0 + 6x$,

$= 6x$,

Q 5 CONT....)

5 MARKS

1 for { radius
circumference
of shell as $2\pi(2-x)$

1 for height of $y/(x^2)$

$$\begin{aligned}\Delta V &= 2\pi(2-x)y\Delta x \\ &= 2\pi(2-x)x^2\Delta x \\ \text{VOL} &= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^2 2\pi(2-x)x^2\Delta x\end{aligned}$$

$$= 2\pi \int_1^2 (2-x)x^2 dx.$$

← 1 for expression of
VOLUME

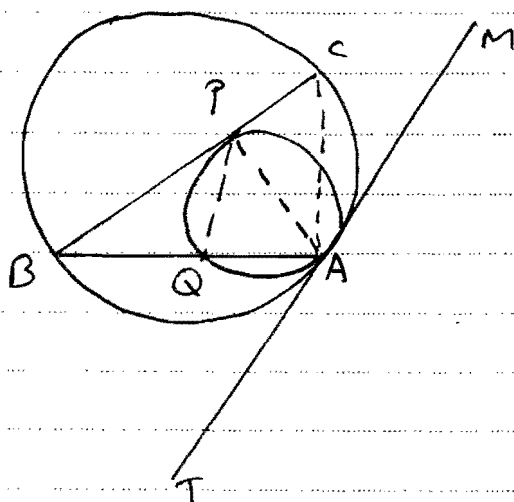
$$= 2\pi \left[\frac{2}{3}x^3 \right]_1^2 - 2\pi \left[\frac{1}{4}x^4 \right]_1^2$$

$$= 2\pi \left(\frac{16}{3} - \frac{2}{3} \right) - 2\pi \left(4 - \frac{1}{4} \right)$$

$$= \frac{28\pi}{3} - \frac{30\pi}{4}$$

$$= \frac{11\pi}{6} \text{ cu units}$$

2 for working
and answer.

QUESTION 6:(a) (i)

(i) NO MARKS

(ii) Taking the tangent TM and the BIG circle

5 MARKS

$$\text{let } \angle MAC = \alpha^\circ$$

1 MARK

$$\therefore \angle CBA = \alpha^\circ \text{ (angle in the alt segment in big circle)}$$

Using the tangent BC and the SMALL circle, let $\angle BPQ = \beta^\circ$

1 MARK

$$\therefore \angle PAQ = \beta^\circ \text{ (angle in the alternate segment)}$$

$$\text{For } \triangle BPQ, \angle PQA = (\alpha + \beta)^\circ \text{ [external angle of } \triangle BPQ \text{]} \quad 1 \text{ MARK}$$

This is the angle in the alternate segment for the small circle and the chord PA with tangent TM.

$$\therefore \angle PAM = (\alpha + \beta)^\circ$$

1 MARK

$$\text{But since } \angle MAC = \alpha$$

$$\therefore \angle PAC = \beta^\circ = \angle PAQ$$

1 MARK

$$\therefore PA \text{ bisects } \angle BAC$$

(subtract 1 if there are no reasons throughout)

Q 6 cont...

$$(b)(i) \quad \mathcal{L} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = 0$$

$$2x/a^2 + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{xb^2}{ya^2}$$

$$\text{At P} \quad m_T = \frac{-a \cos \theta b^2}{a^2 b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

3 MARKS← 1 MARK to here
(can be quoted)At P, tangent is:

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\therefore ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$\therefore ay \sin \theta + bx \cos \theta = ab(\sin^2 \theta + \cos^2 \theta)$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

1 MARK to here

1 for division by ab

(ii) N is the point $(a \cos \theta, 0)$ T is where the tangent cuts $y=0$

$$\text{ie T is } \left(\frac{a}{\cos \theta}, 0 \right)$$

2 MARKS

1 for N

1 for T

$$\therefore ON \cdot OT = a \cos \theta \cdot \frac{a}{\cos \theta}$$

$$= a^2$$

5 MARKS(iii) R is where the tangent meets $x = a/e$

$$\text{ie } \frac{a \cos \theta}{e} + \frac{y \sin \theta}{b} = 1$$

$$\therefore y = \frac{b}{\sin \theta} \left(1 - \frac{\cos \theta}{e} \right)$$

1 for R

$$\text{Slope PS} = m_1 = \frac{b \sin \theta}{a \cos \theta - ae}$$

$$\text{slope RS} = m_2 = \frac{\frac{b}{\sin \theta} \left(1 - \frac{\cos \theta}{e} \right)}{\frac{a}{e} - ae}$$

1 MARK for both

Teacher's Name:

Student's Name/N^o:Q6 cont...

$$\therefore m_1 m_2 = \frac{b \sin \theta}{a(\cos \theta - e)} \times \frac{b}{\sin \theta \left(1 - \frac{\cos \theta}{e}\right)} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \\ \\ 2 \text{ for this} \\ \text{working} \end{array}$$

$$= \frac{b^2/e (e - \cos \theta)}{a^2/e (1 - e^2)(\cos \theta - e)}$$

$$= \frac{b^2}{a^2(1 - e^2)}$$

and since $b^2 = a^2(1 - e^2)$

$$m_1 m_2 = -1 \Rightarrow PS \perp RS$$

$\therefore$ PR subtends a right angle at S

1 for recognising this.

} no special marks
for this.

QUESTION 7:3 MARKS

(a) $x = a \tan \theta$

$$\frac{dx}{d\theta} = a \sec^2 \theta$$

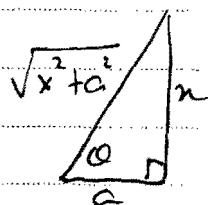
$$\therefore dx = a \sec^2 \theta d\theta$$

$$\therefore \int \frac{dx}{(a^2 + x^2)^{3/2}} = \int \frac{a \sec^2 \theta d\theta}{(a^2 + a^2 \tan^2 \theta)^{3/2}}$$

$$= \int \frac{a \sec^2 \theta d\theta}{a^3 (1 + \tan^2 \theta)}$$

$$= \int \frac{d\theta}{a^2 \sec \theta}$$

$$= \frac{1}{a^2} \sin \theta + k$$



Since $x = a \tan \theta$

$$\therefore \sin \theta = \frac{x}{\sqrt{x^2 + a^2}}$$

$$\therefore \int \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2 + a^2}} + k$$

(b) let the roots of $z^5 = 1$ be the solutions to $\text{cis } 5\theta = 1$

$$\therefore 5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$$\therefore z = 1, \text{cis } \frac{2\pi}{5}, \text{cis } \frac{4\pi}{5}, \text{cis } \frac{6\pi}{5}, \text{cis } \frac{8\pi}{5}$$

(i)

$$(2) \quad w_1 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$w_4 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

$$= \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$$

$$= \bar{w}_1$$

Similarly $w_3 = \bar{w}_2$

1 for this or equivalent

1 to get here.

1 for $\sin \theta$

[NO PENALTY for no k]

3 MARKS

1 MARK.

It is OK here to say "similarly".

Q 7 cont...

(β) Sum of roots of $z^5 - 1 = 0$

1 mark

$$\therefore 1 + w_1 + w_2 + w_3 + w_4 = 0$$

$$\therefore w_1 + w_2 + w_3 + w_4 = -1$$

$$(γ) w_1 + w_2 + w_3 + w_4 = -1$$

$$\therefore w_1 + w_2 + \bar{w}_2 + \bar{w}_1 = -1$$

} 1 mark

$$\therefore 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = -1$$

$$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

(ii) Sum of roots in pairs is

$$w_1 + w_2 + w_3 + w_4 + w_1 w_2 + w_1 w_3 + w_1 w_4 + w_2 w_3 + w_2 w_4 + w_3 w_4 = 0$$

$$\therefore -1 + w_1 \bar{w}_1 + w_2 \bar{w}_2 + w_1 \bar{w}_2 + w_2 \bar{w}_1 + w_1 \bar{w}_3 + w_3 \bar{w}_1 + w_1 \bar{w}_4 + w_4 \bar{w}_1 + w_2 \bar{w}_3 + w_3 \bar{w}_2 + w_2 \bar{w}_4 + w_4 \bar{w}_2 + w_3 \bar{w}_4 + w_4 \bar{w}_3 = 0$$

$$\therefore -1 + w_1(w_2 + \bar{w}_2) + |w_1|^2 + |w_2|^2 + \bar{w}_1(w_2 + \bar{w}_2) = 0$$

$$\therefore -1 + (w_2 + \bar{w}_2)(w_1 + \bar{w}_1) + 2 = 0$$

$$\therefore 2 \cos \frac{4\pi}{5} \cdot 2 \cos \frac{2\pi}{5} + 1 = 0$$

$$\therefore 4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} + 1 = 0$$

} 1 for using $|w|^2 = 1$
1 for using conjugates.

(iii) If the roots of the quadratic are $\cos \frac{2\pi}{5}$ and $\cos \frac{4\pi}{5}$ then

$$\text{Sum} = \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

$$\text{Product} = \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -\frac{1}{4}$$

} 1 for seeing this

 $\therefore$ Quadratic is

$$x^2 + \frac{1}{2}x - \frac{1}{4} = 0$$

$$\therefore 4x^2 + 2x - 1 = 0$$

Teacher's Name:

Student's Name/N^o:Q7 cont...

$$(c)(i) I_n = \int \sec^n \theta d\theta$$

$$= \int \sec^2 \theta \sec^{n-2} \theta d\theta$$

4 MARKS

$$= \tan \theta \sec^{n-2} \theta - \int \tan \theta \frac{d}{d\theta} \sec^{n-2} \theta d\theta$$

1 MARK

Now $\frac{d}{d\theta} \sec^{n-2} \theta = (n-2) \sec^{n-3} \theta (-1) \cos^2 \theta (-\sin \theta)$

$$= (n-2) \sec^{n-2} \theta \tan \theta$$

1 for the differentiation of $\sec \theta$

$$\therefore \int \tan \theta \frac{d}{d\theta} \sec^{n-2} \theta d\theta$$

$$= (n-2) \int \tan^2 \theta \sec^{n-2} \theta d\theta$$

$$= (n-2) \int (\sec^2 \theta - 1) \sec^{n-2} \theta d\theta$$

1 for changing $\tan^2 \theta$ to $(\sec^2 \theta - 1)$

$$\therefore I_n = \tan \theta \sec^{n-2} \theta - (n-2) \int \sec^2 \theta d\theta + (n-2) \int \sec^{n-2} \theta d\theta$$

$$= \tan \theta \sec^{n-2} \theta - (n-2) I_n + (n-2) I_{n-2}$$

Using limits.

$$\therefore (n-1) I_n = \tan \theta \sec^{n-2} \theta \Big|_0^{\pi/4} + (n-2) I_{n-2}$$

1 for evaluating the limits

$$\therefore (n-1) I_n - (n-2) I_{n-2} = (\sqrt{2})^{n-2}$$

(ii) $\therefore \int_0^{\pi/4} \sec^4 \theta d\theta$ means $n=4$.

2 MARKS

$$\therefore 3I_4 - 2I_{n-2} = (\sqrt{2})^2$$

1 for this

$$\therefore 3I_4 = 2 + 2 \int_0^{\pi/4} \sec^2 \theta d\theta$$

$$= 2 + 2 [\tan \theta]_0^{\pi/4}$$

1 for this

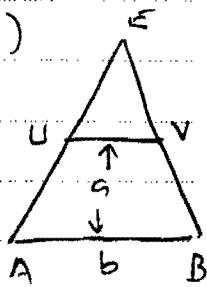
$$\pi/4 = 1$$

$$\therefore \int_0^{\pi/4} \sec^4 \theta d\theta = 4/3$$

NO PARTICULAR MARK

QUESTION 8:

(o)(i)



By similarity

$$\frac{UV}{b} = \frac{p-a}{p}$$

$$\therefore UV = \frac{b(p-a)}{p}$$

$$\therefore (\text{slice}) \triangle V = \frac{b(p-a)}{p} \cdot h \cdot \Delta a$$

3 MARKS

2 for getting the width UV.

1 MARK

$$(ii) \quad Vol = \lim_{\Delta a \rightarrow 0} \sum_0^p \frac{b(p-a)}{p} h \Delta a$$

2 MARKS

$$= \int_0^p \left(hb - \frac{bah}{p} \right) da$$

$$= hba \Big|_0^p - \frac{bah^2}{2p} \Big|_0^p$$

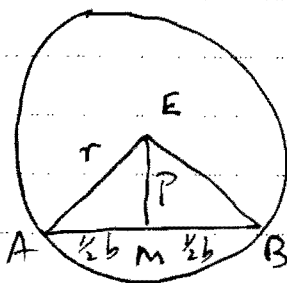
$$= hbp - \frac{bhp}{2}$$

$$= \frac{1}{2} bhp$$

← 1 for this

1 for completion

(iii)



If $\angle AEB = \frac{2\pi}{n}$, there are n of the triangular prisms in the cylinder.

As $n \rightarrow \infty$, $\angle AEB \rightarrow 0$ and $p \rightarrow r$

5 MARKS

1 for seeing this

So volume of cylinder is

$$\lim_{n \rightarrow \infty} nV \text{ where } V \text{ is from (ii) above}$$

Now In $\triangle EMB$, $\tan \frac{\pi}{n} = \frac{\frac{1}{2}b}{p}$

As $n \rightarrow \infty$, $\frac{\pi}{n} \rightarrow 0$, so $\tan \left(\frac{\pi}{n} \right) \rightarrow \frac{\pi}{n}$ (given) 1 for using this fact

$$\therefore \frac{\frac{1}{2}b}{p} \rightarrow \frac{\pi}{n}$$

$$\text{ie. } b \rightarrow \frac{2\pi p}{n}$$

1 for getting b.

Q8 (c) cont...

$$\text{Since } V_{\text{cylinder}} = \lim_{n \rightarrow \infty} nV$$

$$= \lim_{n \rightarrow \infty} n \left(\frac{1}{2} p b h \right)$$

$$= n \left(\frac{\pi p^2 h}{n} \right)$$

$$= \pi p^2 h$$

$$= \pi r^2 h \left[\begin{array}{l} \text{since} \\ p \rightarrow r \end{array} \right]$$

1 for simplification

1 for $p \rightarrow r$

(c)

$$\frac{d^2 x}{dt^2} = -n^2 x \text{ and } \frac{d^2 y}{dt^2} = -n^2 y$$

6 marks

$$\Rightarrow x = a \cos(nt + \alpha) \quad y = b \cos(nt + \beta)$$

$$\therefore \dot{x} = -a n \sin(nt + \alpha) \quad \dot{y} = -b n \sin(nt + \beta)$$

1 for both of these

$$\text{At } t=0, \frac{dx}{dt} = 0$$

$$\text{At } t=0, \dot{y} = 3n$$

$$\therefore 3n = -b n \sin \beta$$

$$\therefore \underline{\alpha = 0}$$

$$\therefore \sin \beta = -\frac{3}{b}$$

1 for α

$$\text{ie } \underline{\dot{x} = -a n \sin(nt)}$$

$$\text{Also at } t=0, y=0$$

$$\text{Also at } t=0, x=4$$

$$4 = a$$

$$\therefore 0 = b \cos \beta$$

1 for a

$$\therefore \underline{\beta = \frac{\pi}{2}}$$

1 for β

$$\therefore x = 4 \cos nt$$

$$\text{Since } \sin \beta = -\frac{3}{b}$$

$$b = -3$$

1 for b

$$\therefore y = 3 \sin(nt + \frac{\pi}{2})$$

$$= 3 \sin(nt)$$

$$\therefore \frac{x^2}{16} + \frac{y^2}{9} = \frac{16 \cos^2 nt}{16} + \frac{9 \sin^2 nt}{9}$$

$$= 1$$

1 for finishing